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OMB No. 0704-0188

REQUIREMENT

1. MATERIAL

   SLEEVE AND SERRATED PIN - ALLOY STEEL PER ASTM-A-331

   MECHANICAL LOCK: CARBON STEEL OR ALLOY STEEL CONFORMING TO THE CHEMICAL COMPOSITION OF 1010, 1020, 1144, 4037, OR 4130 OF FEDERAL STANDARD 68 OR A286 CRES CONFORMING TO THE CHEMICAL LIMITS OF AMS 5737, OR TYPE 316 CRES CONFORMING TO THE CHEMICAL REQUIREMENTS OF AMS 5690

2. HARDNESS

   SLEEVE: KNOOP (1000 GRAM LOAD) 328-468

   SERRATED PIN: ROCKWELL "C" 48 TO 53

   MECHANICAL LOCK: NOT APPLICABLE

3. PROTECTIVE TREATMENT

   SLEEVE: CADMIUM PLATE PER QQ-P-416, TYPE II, CLASS 2 OR ALUMINUM COATING PER MIL-C-83488.

   SERRATED PIN: CADMIUM PLATE PER QQ-P-416, TYPE I, CLASS 3 WITH HYDROGEN EMBRITTLEMENT RELIEF PER MIL-F-81177 OR ALUMINUM COATING PER MIL-C-83488.

   MECHANICAL LOCK: CADMIUM PLATE PER QQ-P-416, TYPE II, CLASS 2 (CARBON AND ALLOY STEEL) PASSIVATE PER QQ-P-35 (A286 AND TYPE 316 CRES) OR ALUMINUM COATING PER MIL-C-83488, AT MANUFACTURER'S OPTION.

4. INSPECTION: DUE TO INSTALLATION LOADS BEING HIGHER THAN PERFORMANCE LOAD REQUIREMENTS, MAGNETIC PARTICLE INSPECTION IS NOT REQUIRED

5. LUBRICATION: NONE, EXCEPT MANUFACTURER'S OPTION TO LUBRICATE LOCK MECHANISM WITH LOX COMPATIBLE DRY FILM LUBRICANTION

6. EXAMPLE OF PART NUMBER

MS 90353 - 0806 D - 1/4 DIA. NOM SIZE, .156 MIN. GRIP, INSTALLED WITH A SINGLE ACTION TOOL

- "D" DENOTES ALUMINUM COATING, NO CODE DENOTES CADMIUM PLATING.
- GRIP RANGE (SEE TABLE II)
- SIZE DASH NUMBER (SEE TABLE I)
- "-" DENOTES FASTENERS INSTALLED WITH A DOUBLE ACTION TOOL, FOR SIZES 05 AND 06
- "-" DENOTES FASTENERS INSTALLED WITH A SINGLE ACTION TOOL, FOR SIZES 08 10, 12, 14, AND 16.
- "S" DENOTES FASTENERS INSTALLED WITH A SINGLE ACTION TOOL, FOR SIZES 05 AND 06.
- "U" DENOTES SINGLE ACTION FASTENERS INSTALLED WITH A SINGLE ACTION TOOL, FOR ALL SIZES (SEE SHEET 1, FOOTNOTE (d))
- MS NUMBER

NOTES:

1. DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCES: DECIMALS ± .010
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS DOCUMENT AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS DOCUMENT SHALL TAKE PRECEDENCE.
3. UNLESS OTHERWISE SPECIFIED, ISSUES OF REFERENCED DOCUMENTS ARE THOSE IN EFFECT AT THE TIME OF SOLICITATION

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY- AV NAVY- AS AIR FORCE- 99 DLA- IS REVIEW ACTIVITIES: MI, 82 USER ACTIVITIES: PROJECT NUMBER: 5320-0815	MILITARY SPECIFICATION SHEET TITLE FASTENER, BLIND, HIGH STRENGTH, PULL TYPE, POSITIVE MECHANICAL LOCK, 100° FLUSH HEAD, ALLOY STEEL, 112 K.S.I Fsu	SPECIFICATION SHEET NUMBER MS90353 28 FEB 1994 REV U SUPERSEDING MS90353 T 13 AUG 1993 AMSC- N/A FSC 5320

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TABLE II. GRIP NUMBER TABULATION (CONTINUED)

GRIP NUMBER SUFFIX	#3/8 (-12)				#7/16 (-14)				#1/2 (-16)			
	GRIP RANGE		C (U)	L	GRIP RANGE		C (U)	L	GRIP RANGE		C (U)	L
	MIN	MAX	MAX	REF	MIN	MAX	MAX	REF	MIN	MAX	MAX	REF
-1200	.219	.282	.820	.548	.281	.346	.944	.662	.344	.407	1.147	.815
-1201	.281	.345	.883	.610	.344	.407	1.026	.724	.406	.470	1.209	.878
-1202	.344	.407	.945	.673	.406	.470	1.089	.787	.469	.532	1.272	.940
-1203	.406	.470	1.007	.735	.469	.532	1.151	.849				
-1204	.469	.532	1.070	.798								
-1205												
-1206												
-1207												
-1208												
-1209	.531	.595	1.133	.860	.531	.595	1.214	.912	.531	.595	1.334	1.003
-1210	.594	.657	1.195	.923	.594	.657	1.276	.974	.594	.657	1.397	1.065
-1211	.656	.720	1.257	.985	.656	.720	1.339	1.037	.656	.720	1.459	1.128
-1212	.719	.782	1.320	1.048	.719	.782	1.401	1.099	.719	.782	1.522	1.190
-1213	.781	.845	1.383	1.110	.781	.845	1.464	1.162	.781	.845	1.584	1.253
-1214	.844	.907	1.445	1.173	.844	.907	1.526	1.224	.844	.907	1.647	1.315
-1215	.906	.970	1.507	1.235	.906	.970	1.588	1.287	.906	.970	1.709	1.378
-1216	.968	1.032	1.570	1.298	.968	1.032	1.651	1.349	.969	1.032	1.772	1.440
-1217	1.031	1.095	1.633	1.360	1.031	1.095	1.714	1.412	1.031	1.095	1.834	1.503
-1218	1.094	1.157	1.695	1.423	1.094	1.157	1.776	1.474	1.094	1.157	1.897	1.565
-1219	1.156	1.220	1.757	1.485	1.156	1.220	1.839	1.537	1.156	1.220	1.959	1.628
-1220	1.219	1.282	1.820	1.548	1.219	1.282	1.901	1.599	1.219	1.282	2.022	1.690
-1221	1.281	1.345	1.883	1.610	1.281	1.345	1.964	1.662	1.281	1.345	2.084	1.753
-1222	1.344	1.407	1.945	1.673	1.344	1.407	2.026	1.724	1.344	1.407	2.147	1.815
-1223	1.406	1.470	2.007	1.735	1.406	1.470	2.088	1.787	1.406	1.470	2.208	1.878
-1224	1.469	1.532	2.070	1.798	1.469	1.532	2.151	1.849	1.469	1.532	2.272	1.940
-1225	1.531	1.595	2.133	1.860	1.531	1.595	2.214	1.912	1.531	1.595	2.334	2.003
-1226	1.594	1.657	2.195	1.923	1.594	1.657	2.276	1.974	1.594	1.657	2.397	2.065
-1227	1.656	1.720	2.257	1.985	1.656	1.720	2.339	2.037	1.656	1.720	2.459	2.128
-1228	1.719	1.782	2.320	2.048	1.719	1.782	2.401	2.099	1.719	1.782	2.522	2.190
-1229	1.781	1.845	2.383	2.110	1.781	1.845	2.464	2.162	1.781	1.845	2.584	2.253
-1230	1.844	1.907	2.446	2.173	1.844	1.907	2.526	2.224	1.844	1.907	2.647	2.315
-1231	1.906	1.970	2.508	2.235	1.906	1.970	2.589	2.287	1.906	1.970	2.709	2.378
-1232	1.969	2.032	2.571	2.298	1.969	2.032	2.651	2.349	1.969	2.032	2.772	2.440
-1233	2.031	2.095	2.633	2.360	2.031	2.095	2.714	2.412	2.031	2.095	2.834	2.503
-1234	2.094	2.157	2.696	2.423	2.094	2.157	2.776	2.474	2.094	2.157	2.897	2.565
-1235	2.156	2.220	2.758	2.485	2.156	2.220	2.839	2.537	2.156	2.220	2.959	2.628
-1236	2.219	2.282	2.820	2.548	2.219	2.282	2.901	2.599	2.219	2.282	3.022	2.690
-1237									2.281	2.345	3.084	2.753
-1238									2.344	2.407	3.147	2.815
-1239									2.406	2.470	3.208	2.878
-1240									2.469	2.532	3.272	2.940

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY-AV NAVY-AS AIR FORCE-99 DLA-IS REVIEW: ML 82 USER: PROJECT NUMBER: 5320-0815	MILITARY SPECIFICATION SHEET TITLE FASTENER, BLIND, HIGH STRENGTH, PULL TYPE, POSITIVE MECHANICAL LOCK, 100° FLUSH HEAD, ALLOY STEEL, 112 K.S.I. Fsu	SPECIFICATION SHEET NUMBER MS90353 REV U 28 FEB 1994 SUPERSEDING MS90353 T 13 AUG 1983 AMSC- N/A FSC 5320
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TABLE II. GRIP NUMBER TERMINATION

GRIP DASH NO.	#5/32 (-05) GRIP RANGE MIN	#5/32 (-05) GRIP RANGE MAX	#5/32 (-05) C (D) MAX	#5/32 (-05) L REF	#3/16 (-06) GRIP DASH NO.	#3/16 (-06) GRIP RANGE MIN	#3/16 (-06) GRIP RANGE MAX	#3/16 (-06) C (D) MAX	#3/16 (-06) L REF	#1/4 (-08) GRIP DASH NO.	#1/4 (-08) GRIP RANGE MIN	#1/4 (-08) GRIP RANGE MAX	#1/4 (-08) C (D) MAX	#1/4 (-08) L REF	#5/16 (-10) GRIP DASH NO.	#5/16 (-10) GRIP RANGE MIN	#5/16 (-10) GRIP RANGE MAX	#5/16 (-10) C (D) MAX	#5/16 (-10) L REF
-0502	.094	.157	.434	.363	-0602	.094	.157	.376	.363	-0802	.094	.157	.376	.406	-1002	.094	.157	.376	.506
-0503	.156	.282	.488	.432	-0603	.156	.282	.438	.432	-0803	.156	.282	.438	.469	-1003	.156	.282	.438	.569
-0504	.219	.345	.551	.494	-0604	.219	.345	.501	.494	-0804	.219	.345	.501	.531	-1004	.219	.345	.501	.631
-0505	.281	.407	.613	.557	-0605	.281	.407	.563	.557	-0805	.281	.407	.563	.594	-1005	.281	.407	.563	.694
-0506	.344	.470	.676	.619	-0606	.344	.470	.626	.619	-0806	.344	.470	.626	.656	-1006	.344	.470	.626	.756
-0507	.406	.532	.738	.682	-0607	.406	.532	.688	.682	-0807	.406	.532	.688	.719	-1007	.406	.532	.688	.818
-0508	.469	.595	.801	.744	-0608	.469	.595	.751	.744	-0808	.469	.595	.751	.781	-1008	.469	.595	.751	.881
-0509	.531	.657	.863	.807	-0609	.531	.657	.813	.807	-0809	.531	.657	.813	.844	-1009	.531	.657	.813	.943
-0510	.594	.720	.926	.869	-0610	.594	.720	.876	.869	-0810	.594	.720	.876	.906	-1010	.594	.720	.876	1.006
-0511	.656	.782	.988	.932	-0611	.656	.782	.938	.932	-0811	.656	.782	.938	.969	-1011	.656	.782	.938	1.068
-0512	.719	.845	1.051	.994	-0612	.719	.845	1.001	.994	-0812	.719	.845	1.001	1.031	-1012	.719	.845	1.001	1.131
-0513	.781	.907	1.113	1.057	-0613	.781	.907	1.063	1.057	-0813	.781	.907	1.063	1.094	-1013	.781	.907	1.063	1.193
-0514	.844	.970	1.176	1.119	-0614	.844	.970	1.126	1.119	-0814	.844	.970	1.126	1.156	-1014	.844	.970	1.126	1.256
-0515	.906	1.032	1.238	1.182	-0615	.906	1.032	1.188	1.182	-0815	.906	1.032	1.188	1.219	-1015	.906	1.032	1.188	1.318
-0516	.969	1.095	1.301	1.244	-0616	.969	1.095	1.251	1.244	-0816	.969	1.095	1.251	1.281	-1016	.969	1.095	1.251	1.381
-0517	1.031	1.157	1.363	1.307	-0617	1.031	1.157	1.313	1.307	-0817	1.031	1.157	1.313	1.344	-1017	1.031	1.157	1.313	1.443
-0518	1.094	1.220	1.426	1.369	-0618	1.094	1.220	1.376	1.369	-0818	1.094	1.220	1.376	1.406	-1018	1.094	1.220	1.376	1.506
-0519	1.156	1.282	1.488	1.432	-0619	1.156	1.282	1.438	1.432	-0819	1.156	1.282	1.438	1.469	-1019	1.156	1.282	1.438	1.568
-0520	1.219	1.345	1.551	1.494	-0620	1.219	1.345	1.501	1.494	-0820	1.219	1.345	1.501	1.531	-1020	1.219	1.345	1.501	1.631
-0521	1.281	1.407	1.613	1.557	-0621	1.281	1.407	1.563	1.557	-0821	1.281	1.407	1.563	1.594	-1021	1.281	1.407	1.563	1.693
-0522	1.344	1.470	1.676	1.619	-0622	1.344	1.470	1.626	1.619	-0822	1.344	1.470	1.626	1.656	-1022	1.344	1.470	1.626	1.756
-0523	1.406	1.532	1.738	1.682	-0623	1.406	1.532	1.688	1.682	-0823	1.406	1.532	1.688	1.719	-1023	1.406	1.532	1.688	1.818
-0524	1.469	1.595	1.801	1.744	-0624	1.469	1.595	1.751	1.744	-0824	1.469	1.595	1.751	1.781	-1024	1.469	1.595	1.751	1.881
-0525	1.531	1.657	1.863	1.807	-0625	1.531	1.657	1.813	1.807	-0825	1.531	1.657	1.813	1.844	-1025	1.531	1.657	1.813	1.943
-0526	1.594	1.720	1.926	1.869	-0626	1.594	1.720	1.876	1.869	-0826	1.594	1.720	1.876	1.906	-1026	1.594	1.720	1.876	2.006
-0527	1.656	1.782	1.988	1.932	-0627	1.656	1.782	1.938	1.932	-0827	1.656	1.782	1.938	1.969	-1027	1.656	1.782	1.938	2.068
-0528	1.719	1.845	2.051	1.994	-0628	1.719	1.845	2.001	1.994	-0828	1.719	1.845	2.001	2.031	-1028	1.719	1.845	2.001	2.131
-0529	1.781	1.907	2.113	2.057	-0629	1.781	1.907	2.063	2.057	-0829	1.781	1.907	2.063	2.094	-1029	1.781	1.907	2.063	2.193
-0530	1.844	1.970	2.176	2.119	-0630	1.844	1.970	2.126	2.119	-0830	1.844	1.970	2.126	2.156	-1030	1.844	1.970	2.126	2.256
-0531	1.906	2.032	2.238	2.182	-0631	1.906	2.032	2.188	2.182	-0831	1.906	2.032	2.188	2.219	-1031	1.906	2.032	2.188	2.318
-0532	1.969	2.095	2.301	2.244	-0632	1.969	2.095	2.251	2.244	-0832	1.969	2.095	2.251	2.281	-1032	1.969	2.095	2.251	2.381

PREPARING ACTIVITY NAVY-AS CUSTODIANS ARMY- AV NAVY- AS AIR FORCE- 99 DLA- IS REVIEW MI, 99 USER: PROJECT NUMBER 5320-0815	MILITARY SPECIFICATION SHEET TITLE FASTENER, BLIND, HIGH STRENGTH, PULL TYPE, POSITIVE MECHANICAL LOCK, 100° FLUSH HEAD, ALLOY STEEL, 112 K.S.I. Fsu	SPECIFICATION SHEET NUMBER MS90353 REV U 28 FEB 94 SUPERSEDING MS90353 T 13 AUG 1993 AMSC- N/A FSC 5320
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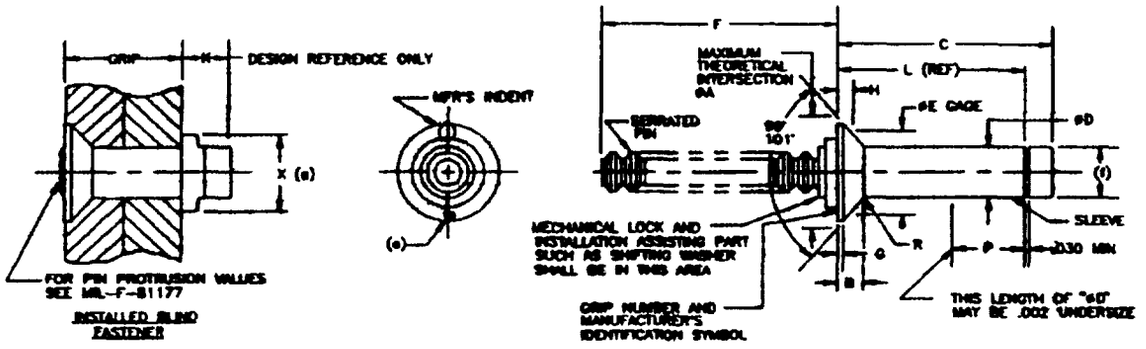


TABLE I. FASTENER DIMENSIONS AND STRENGTHS

(b) SIZE DASH NUMBER			NOM. SIZE	øA	B	øD	øE	F	G	H	K	P	R	X	INSTALLED FASTENER		HOLE LIMITS
DOUBLE ACTION (c)	SINGLE ACTION (d)	SINGLE ACTION (d)			MAX.		GAGE	MIN.	MAX.	GAGE	MAX.	MAX	RAD MAX.	(e) MIN	MINIMUM SINGLE SHEAR LBS.	MINIMUM TENSILE LBS.	
~~-05~~	S05	U05	5/32	.333	.072	.164 .162	.3832 .3830	.844	.011	.0208 .0171	.202	.215	010	.195	2340	1350	.164 .167
~~-06~~	S06	U06	3/16	.396	.090	.199 .197	.3272 .3270	.875	.013	.0243 .0208	.231	.250	015	.238	3450	2100	.199 .202
-	-08	U08	1/4	.507	.105	.260 .258	.4330 .4318	1.000	.017	.0310 .0270	.279	.305	.020	.315	5800	3650	.260 .263
-	-10	U10	5/16	.634	.137	.312 .310	.5451 .5449	1.218	.020	.0387 .0325	.319	.350	025	.373	8500	5200	.312 .315
-	-12	U12	3/8	.762	.168	.374 .372	.6982 .6980	1.598	.023	.0438 .0388	.384	.406	030	.448	12280	7500	.374 .377
-	-14	U14	7/16	.880	.185	.437 .435	.7984 .7982	1.842	.025	.0488 .0468	.419	.480	.032	.522	16700	10150	.437 .441
-	-16	U16	1/2	1.017	.229	.498 .496	.8902 .8900	1.842	.028	.0622 .0456	.540	.598	.035	.598	21800	13500	.500 .504

(a) MAXIMUM DIAMETER OF BLIND HEAD (X) SHALL NOT EXCEED MAXIMUM DIAMETER OF MANUFACTURED HEAD (A).

(b) SIZES "05" AND "06" WITH "-" ARE INSTALLED WITH A DOUBLE ACTION TOOL. ALL OTHER SIZES ARE INSTALLED WITH A SINGLE ACTION TOOL. SIZES "U05" AND "U06" MAY BE INSTALLED WITH A DOUBLE OR SINGLE ACTION TOOL. SEE MIL-F-81177, APPENDIX A FOR TOOL SELECTION CHART.

(c) ~~-05, -06, S05, S06, -08, -10, -12, -14 AND -16~~ ARE INACTIVE FOR NEW DESIGN AFTER 1 JAN 91 USE U05, U06, S05 AND S06.

(d) LETTER "U" SIGNIFIES SINGLE ACTION FASTENER WITHOUT PAINTABLE TYPE OF LOCKING COLLAR OR LOCKING COLLAR CAP. "U" AND "S" CODE FASTENERS IN SIZES "05" AND "06" ARE COLOR IDENTIFIED GREEN ON LOCKING COLLAR, SHIFTING WASHER, OR SERRATED PIN. "S" CODE FASTENERS MAY BE SUPPLIED WITH A LOCKING COLLAR CAP.

(e) SINGLE DIGIT MARKING IS PERMISSIBLE FOR GRIP DASH NUMBERS LESS THAN 10.

(f) DIAMETER OF SERRATED PIN HEAD SHALL NOT EXCEED "øD" MAX AND MAY BE .002 BELOW "øD" MIN. IN AREA "P" ONLY.

(U) DENOTES CHANGE(S)

INCH-POUND

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY-AV NAVY-AS AIR FORCE-99 DLA-IS REVIEW USER: ML 82 PROJECT NUMBER 5320-0815	MILITARY SPECIFICATION SHEET TITLE FASTENER, BLIND, HIGH STRENGTH, PULL TYPE, POSITIVE MECHANICAL LOCK, 100° FLUSH HEAD, ALLOY STEEL, 112 K.S.I. Fsu	SPECIFICATION SHEET NUMBER MS90353 28 FEB 94 REV U SUPERSEDING MS90353 T 13 AUG 1993 AMSC- N/A FSC 5320
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